

Deterministic Optimization

Robust Optimization

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Concept of Robustness in Example

Robust Optimization

Learning Objectives for this module

- Understand the concept of robustness in the drug production example

Example: Drug Production

- A company produces two kinds of drugs, Drug I and Drug II, containing a specific active agent A, which is extracted from two kinds of raw materials, Raw I and Raw II, purchased on the market.
- Production data:
 - Selling prices of Drug I and Drug II (\$/1000 packs): 6200, 6900
 - Content of agent A in Drug I and Drug II (g/kg): 0.5, 0.6
 - Man power required (hours/1000 packs): 90.0, 100.0
 - Equipment required (hours/1000 packs): 40.0, 50.0
 - Operational cost (\$/1000 packs): 700, 800
 - Purchasing price of Raw I and Raw II (\$/kg): 100.00, 199.90
 - Content of agent A in Raw I and Raw II (g/kg): 0.01, 0.02

Example: Drug Production

- Resource constraints:
 - Budget (\$): 100,000
 - Man power (hour): 2,000
 - Equipment (hour): 800
 - Capacity of raw material storage (kg): 1,000
- Objective: find the production plan that maximizes the profit of the company
- We can easily formulate a linear programming problem

Example: Drug Production

	Purchasing and operation cost	Revenue
min	$100 \cdot RawI + 199.9 \cdot RawII + 700 \cdot DrugI + 800 \cdot DrugII$	$-(6200 \cdot DrugI + 6900 \cdot DrugII)$
s.t.	$0.01 \cdot RawI + 0.02 \cdot RawII - 0.5 \cdot DrugI - 0.6 \cdot DrugII \geq 0$	← Balance of agent A
	$RawI + RawII \leq 1000$	← Storage constraint
	$90 \cdot DrugI + 100 \cdot DrugII \leq 2000$	← Manpower constraint
	$40 \cdot DrugI + 50 \cdot DrugII \leq 800$	← Equipment constraint
	$100 \cdot RawI + 199.9 \cdot RawII + 700 \cdot DrugI + 800 \cdot DrugII \leq 100000$	← Budget constraint
	$RawI, RawII, DrugI, DrugII \geq 0.$	← Nonnegativity constraint

- Optimal solution: $RawI = 0$, $RawII = 438.789$, $DrugI = 17.552$, $DrugII = 0$
- Optimal objective: -8819.658

Impact on Production and Profit

- Optimal production:
 - $RawI = 0$, $RawII = 438.789$, $DrugI = 17.552$, $DrugII = 0$
- With 50% chance, content of active agent in $RawII$ is 0.0196g/kg, this results in $438.789 \cdot 0.0196 = 8.6003$ g active agents
- But to produce 17.552 units of $DrugI$ needs $0.5 \cdot 17.552 = 8.776$ g active agent
- The optimal production plan becomes infeasible!
- Say in this case the company adapts to the low content of active agent in raw material and cuts production to $\frac{8.6003}{0.5} = 17.201$
- In this case, the profit is $17.201 \cdot (6200 - 700) - 199.9 \cdot 438.789 = 6891.6$

The Unreliability of Nominal Solution

- In the drug company example, a 2% change in the active agent data will cause a 21% drop in profit with 50% chance, and a 11% drop in profit on average!
- The source of the problem is that the company only plans according to the nominal value of the uncertain data, which produces a nominal solution that is very unreliable
- The way to overcome this issue is the concept of robustness

The Concept of Robustness

- We want the production plan to survive “all realizations” of the content of active agents:
 - The solution ($RawI$, $RawII$, $DrugI$, $DrugII$) should satisfy

$$activeI \cdot RawI + activeII \cdot RawII - 0.5 \cdot DrugI - 0.6 \cdot DrugII \geq 0$$

for all $activeI \in [0.00995, 0.01005]$, $activeII \in [0.0196, 0.0204]$

- This constraint is called a **Robust Constraint**
- It's equivalent to a usual deterministic constraint using “worst case”:

$$0.00995 \cdot RawI + 0.0196 \cdot RawII - 0.5 \cdot DrugI - 0.6 \cdot DrugII \geq 0$$

Robust Drug Production Problem

- Incorporating the robust constraint, we have a robust drug production problem

$$\min \quad 100 \cdot \text{RawI} + 199.9 \cdot \text{RawII} + 700 \cdot \text{DrugI} + 800 \cdot \text{DrugII} - (6200 \cdot \text{DrugI} + 6900 \cdot \text{DrugII})$$

$$\text{s.t.} \quad 0.00995 \cdot \text{RawI} + 0.00196 \cdot \text{RawII} - 0.5 \cdot \text{DrugI} - 0.6 \cdot \text{DrugII} \geq 0$$

$$\text{RawI} + \text{RawII} \leq 1000$$

$$90 \cdot \text{DrugI} + 100 \cdot \text{DrugII} \leq 2000$$

$$40 \cdot \text{DrugI} + 50 \cdot \text{DrugII} \leq 800$$

$$100 \cdot \text{RawI} + 199.9 \cdot \text{RawII} + 700 \cdot \text{DrugI} + 800 \cdot \text{DrugII} \leq 100000$$

$$\text{RawI}, \text{RawII}, \text{DrugI}, \text{DrugII} \geq 0.$$

- Robust optimal solution: RawI = 877.732, RawII = 0, DrugI = 17.467, Drug II = 0, OPT = -8294.567

Summary

- We learned:
 - Concept of robustness
 - A simple robust constraint